

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Eighth Semester of B. Tech. (EE) Examination

May 2014

EE-416 Advanced Microcontrollers & Applications

Date: 24.05.2014, Saturday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 State various peripherals available in c8051f120 microcontroller along with their specification details [07]

Q - 2 (a) Attain the Following for c8051f120. [06]

1. How internally 98 MHz System Clock can be generated?
2. Calculate the Resolution for ADC0.
3. Calculate the Overflow Time for Timer0 in 16 bit mode if Clock frequency to Timer Module is configured to be 98 MHz/12.
4. Write down the contain of P0MDOUT to configure the Port0 in Push pull mode.
5. Write down Possible configuration of Clock source to timer module of TIMER2.
6. Write down the contain of AMX0SL & AMX0CF to configure first channel of ADC0 in Bipolar mode.

(b) Explain various control loops used in C programming of Microcontroller. [08]

OR

(a) Why Programs for Microcontroller uses infinite loop? Explain the same with suitable example. Also write program to put microcontroller in infinite loop with use of any two control loops. [06]

(b) Explain any 2 keywords used in C programming of Microcontroller on Keil Compiler Platform. [04]

- (c) Write the various types of variable in C Programming along the details of Memory it occupies in bytes. [04]

Q – 3 (a) Write a Program to toggle Port pins on Port 1 using a for loop delay. [06]

(b) Explain the following term with respect to c8051f120 microcontroller, [08]

1. Crossbar & Crossbar Decoder.
2. SFRPAGE Register & SFR Paging.
3. PLL & Name of Associated Register.
4. Push Pull & Open Drain Mode of Port Pin with rough diagram.

OR

Q – 3 (a) Explain the the following terms. [06]

1. Sourcing & Sinking Current of Device.
2. MIPS & MFLOPS
3. Interrupt Latency.

(b) The aim of following program is to toggle the led at a fixed rate in [08]

ISR of timer. Find out all possible error due to which the program may not be compiled or may not work properly as per it's objective. Also write down the correct program. (consider 8051 Microcontroller)

Program:

```
#include<reg51.h>
```

```
void main()
```

```
{ EA=1;
```

```
TMOD=0x02;
```

```
TR0=1;
```

```
while(1){;}
```

```
void timer0_isr()
```

```
{led=!led;}
```


SECTION – II

Q - 4 Explain the DC motor Speed control with the help of Block diagram. If this is to be implemented using **c8051f120** then List out the required SFR for the same. No need to write the Program. [07]

Q -5 (a) Explain the Timer2 in **16 Bit Auto Reload** mode with required block diagram. [07]

(b) Explain working of ADC2 with required Block diagram. [07]

OR

(a) Write a program (Show necessary calculations) to measure voltage available on AIN0.0 with ADC start of conversion through TIMER2 overflow. [07]

(b) Write a program to generate a sine wave signal on DAC0. Assume suitable data. Show necessary Calculations. [07]

Q – 6 (a) Explain working of DAC0 with required block diagram. [04]

(b) Write & explain a program for single phase Over current Relay, which measures equivalent RMS current as a DC Signal available on Pin AIN0.0. Show necessary calculations & Assume suitable datas. [10]

OR

(a) Write & explain a simple program that demonstrates the use of RTOS rtx51TNY available with Keil IDE to toggle the LED connected on Port P0.0. [06]

(b) Explain the concept of Super loop & execution in Round Robbin Task scheduling. Also explain various state of Task in RTOS. [08]
